

# Focus



Watch the video and refer to the link to understand how the colour sensor detects and outputs RGB values.

Observe and record the RGB ranges for at least five colours as shown in the experiment.

Fill in the table with the Red (R), Green (G), and Blue (B) values for each colour.

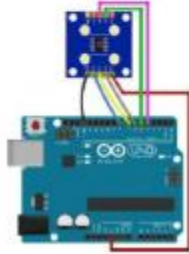
Color Sensor RGB Value Table

| Color  | Red (R) | Green (G) | Blue (B) |
|--------|---------|-----------|----------|
| Red    |         |           |          |
| Green  |         |           |          |
| Blue   |         |           |          |
| Yellow |         |           |          |
| Purple |         |           |          |

# Practice



Let's connect the color sorting sensor to the Arduino.



## Assessment

### Focus Questions:

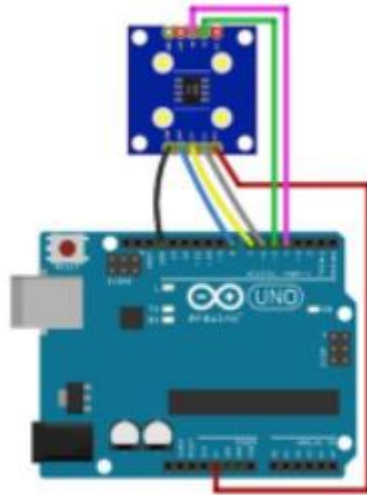
1. *How does the color sensor detect different colors?*
  - **Strategy:** Use a concept map to link sensor components to their functions.
2. *What factors might affect the accuracy of color detection?*
  - **Strategy:** Brainstorm and list variables such as lighting, distance, and sensor calibration.
3. *How can color sensors be used in automation?*
  - **Strategy:** Think-Pair-Share—students discuss their ideas with a partner before sharing with the class.

## Practice – Building and Testing the Color Sensor

### Materials Needed:

- Arduino Uno
- Color sensor module (TCS3200)
- Jumper wires

- Breadboard
- Colored objects for testing
- Computer with Arduino IDE



#### Step-by-Step Instructions:

##### 1. Connect the color sensor to the Arduino:

- VCC → 5V
- GND → GND
- S0 & S1 → Digital Pins (for frequency scaling)
- S2 & S3 → Digital Pins (to select color filters)
- OUT → Digital Pin (to read color frequency)

##### 2. Upload the Arduino Code:

Use a predefined script to read and display RGB values.

```
// Define sensor pins
```

```
#define S0 4
```

```
#define S1 5
```

```
#define S2 6
```

```
#define S3 7
```

```
#define sensorOut 8
```

```
// Variables for color frequency
```

```
int redFrequency = 0;
int greenFrequency = 0;
int blueFrequency = 0;

void setup() {
  pinMode(S0, OUTPUT);
  pinMode(S1, OUTPUT);
  pinMode(S2, OUTPUT);
  pinMode(S3, OUTPUT);
  pinMode(sensorOut, INPUT);

  // Set frequency scaling to 20%
  digitalWrite(S0, HIGH);
  digitalWrite(S1, LOW);

  Serial.begin(9600);
}

void loop() {
  // Read Red value
  digitalWrite(S2, LOW);
  digitalWrite(S3, LOW);
  redFrequency = pulseIn(sensorOut, LOW);

  // Read Green value
  digitalWrite(S2, HIGH);
  digitalWrite(S3, HIGH);
  greenFrequency = pulseIn(sensorOut, LOW);

  // Read Blue value
  digitalWrite(S2, LOW);
  digitalWrite(S3, HIGH);
```

```
blueFrequency = pulseIn(sensorOut, LOW);
```

```
// Print RGB values
```

```
Serial.print("R: ");
```

```
Serial.print(redFrequency);
```

```
Serial.print(" G: ");
```

```
Serial.print(greenFrequency);
```

```
Serial.print(" B: ");
```

```
Serial.println(blueFrequency);
```

```
delay(500); // Small delay for better readability
```

```
}
```

3. **Test with Different Colors:**

- Place objects of different colors in front of the sensor.
- Record the RGB values in the provided table.

4. **Analyze and Interpret the Results:**

- Compare values for different colors.
- Identify trends in RGB detection.

**Reflection Questions:**

1. What patterns did you notice in the RGB values?
2. How can we improve the accuracy of color detection?
3. In what real-life applications could this technology be useful?

## 1. Evaluate:

**Teacher Role:**

- Facilitate assessments to measure students' understanding and practical skills.
- Observe and provide constructive feedback on students' circuit connections, coding, and troubleshooting processes.
- Assess students' ability to apply knowledge in a real-world automation context.
- Use a structured rubric to ensure fair and consistent evaluation.

**Expected From Students:**

- Answer focus questions to demonstrate conceptual understanding.
- Present their Color Sorting Machine project and explain how the color sensor and servo motor work.
- Showcase troubleshooting steps if their system encounters errors.
- Reflect on real-world applications of their work in automation and industry.

### Showcase: Evaluating the Main Project

Students will present their working Color Sorting Machine, demonstrating:

- How the TCS3200 color sensor detects different colors.
- The Arduino code they used and any modifications made.
- Troubleshooting steps they took if the system failed.

### Assessment Rubric for Teachers

| Criteria                                | 4 - Excellent                                                        | 3 - Proficient                           | 2 - Developing                                                 | 1 - Needs Improvement                                |
|-----------------------------------------|----------------------------------------------------------------------|------------------------------------------|----------------------------------------------------------------|------------------------------------------------------|
| <b>Circuit Assembly</b>                 | All components are correctly connected and functional.               | Minor issues, but the system works.      | Some connections are incorrect; requires teacher intervention. | Major wiring errors; system does not function.       |
| <b>Code Implementation</b>              | Code is fully functional and well-structured.                        | Minor issues, but the code works.        | Code has errors; requires debugging help.                      | Code does not work; lacks key components.            |
| <b>Troubleshooting &amp; Debugging</b>  | Identifies and solves problems independently.                        | Identifies issues but needs some help.   | Requires significant guidance to troubleshoot.                 | Unable to diagnose or fix issues.                    |
| <b>Concept Understanding</b>            | Clearly explains how the color sensor and servo motor work together. | Explains most concepts correctly.        | Shows partial understanding; some misconceptions.              | Struggles to explain the concepts.                   |
| <b>Real-World Application</b>           | Connects project to real-world automation with detailed examples.    | Makes general connections to automation. | Identifies real-world relevance but lacks depth.               | Unable to relate project to real-world applications. |
| <b>Presentation &amp; Communication</b> | Confidently presents the project with clear explanations.            | Presents well but lacks some clarity.    | Struggles with explanations and technical terms.               | Limited or unclear explanation of the project.       |

### Relation to Standards

This project aligns with **NGSS (Next Generation Science Standards)** and **ISTE (International Society for Technology in Education)**:

**NGSS - Engineering Design (MS-ETS1-1, MS-ETS1-3)**: Defines the problem and designs a solution using automation.

**NGSS - Physical Science (PS4.C: Electromagnetic Radiation)**: Understanding how the sensor detects colors using light reflection.

**ISTE 4a-4d (Innovative Designer)**: Students use **Arduino programming** and **troubleshooting strategies** to build an automated system.

**ISTE 5a-5d (Computational Thinker)**: Students develop and debug an algorithm for sorting objects by color.

## **Color Sorting Machine**

### **Learning Outcomes**

By the end of the theme student will be able to:

- Select Optimal Cam Type.
- Identify the best belt type for your system.
- Research and apply methods to prevent jams.
- Ensure smooth object flow through the system.
- Identify Belt Drive Components: Recognize pulleys, shafts, and belts.
- Create conveyors and bins for sorting.
- Assemble System: Accurately build the sorting system.
- Create Linkages: Design linkages to control sorting arms or gates.
- Wire Color Sensor: Connect a color sensor to an electrical circuit.
- Control Multiple Servos: Connect and synchronize multiple servo motors.
- Calibrate Sensor: Test and fine-tune the color sensor.
- Power and Sync Servos: Configure the circuit to power and sync servos.
- Learn New Functions: Understand and apply new functions and loops.
- Link Signals: Connect sensor input to servo motor output.
- Integrate Arduino with Python: Interface Arduino with Python for control.

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## Lesson 5

### UNIT 02: [Intermediate]

#### Chapter 01: Robotics

#### Lesson 02: Connect Servos

**Time Frame:** 90 minutes

| Activity Name                                 | Duration | Description                                                                                                                                                                     |
|-----------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Engage: Introduction &amp; Warm-Up</b>     | 10 min   | Start with a quick discussion: <i>“Where do we see color sensors in real life?”</i> Show images or short videos of automated sorting systems (e.g., recycling, conveyor belts). |
| <b>Think: Brainstorming &amp; Predictions</b> | 10 min   | Students predict how a color sensor might work. Ask: <i>“What do you think happens inside the sensor when it detects different colors?”</i>                                     |
| <b>Explore: Hands-on Circuit Building</b>     | 25 min   | Students connect the color sensor to Arduino, following step-by-step guidance. They test it by placing different colored objects in front of the sensor.                        |
| <b>Explain: Watch &amp; Discuss</b>           | 15 min   | Watch the Engineering Mindset video on color sensors. Discuss how the color sensor detects and differentiates colors. Students update their EDP papers with findings.           |
| <b>Elaborate: Coding &amp; Testing</b>        | 20 min   | Students write/upload Arduino code to display detected colors on the serial monitor or an LCD screen. They tweak code parameters for better accuracy.                           |
| <b>Evaluate: Showcase &amp; Feedback</b>      | 10 min   | Each student/team presents their working setup. Use a rubric-based evaluation for feedback. Students reflect on what worked well and what challenges they faced.                |

### 1. Engage:

#### Teacher Role:

- Facilitate discussions by asking thought-provoking questions.
- Encourage students to think about real-world applications of automation.
- Guide students in making connections between color detection and motor movement.



**Expected From Students:**

- Actively participate in discussions and share their ideas.
- Reflect on experiences with automated sorting systems.
- Predict how a servo motor can be used for color-based sorting.

**Intro Story: Automated Factory Conveyor****Strategy: “Imagine Yourself There” (Role-Playing)**

- The teacher sets up a scenario where students become engineers in a high-tech factory.
- The factory needs an automated sorting system that uses a color sensor to detect colors and a servo motor to move objects into the correct bin.
- Students visualize themselves designing this system.

**Teacher Says:**

*"You are now engineers in a high-tech factory. Your job is to create a system that sorts objects by color. The color sensor detects the color, and a motor moves the object to the correct place. But what happens if the motor moves too much? Or too little? Let's figure it out!"*

**Story Reflection****Questions:**

1. Have you ever seen machines that automatically sort objects? Where?
2. How do you think a robot can detect different colors?
3. Why do we need motors in sorting systems?
4. What do you think will happen if the motor doesn't move to the right position?

**Suggested Answers:**

1. *Yes! In supermarkets, luggage sorting at airports, and recycling centers.*
2. *A robot uses color sensors to detect the light reflected from objects.*
3. *Motors move objects to the correct location based on sensor data.*
4. *If the motor moves incorrectly, the object will end up in the wrong bin.*

**Strategy: “Think-Pair-Share”**

- **Think:** Students individually think about their answers.

- **Pair:** They discuss their thoughts with a partner.
- **Share:** Pairs share their ideas with the class, and the teacher connects their responses to real-world automation.

## 2. Think:

### Questions and Possible Answers

#### 1. If the servo motor controls the conveyor belt, what happens if it stops working? How can we design a backup system?

- *Possible Answers:*
  - The candies won't reach the correct sorting area.
  - A secondary motor could take over when the first one fails.
  - Manual intervention or a sensor alert system could notify workers.
  - A battery backup or emergency stop mechanism could be added.

#### 2. The servo motor relies on sensors to guide the candies. What challenges might arise if the sensors fail to detect them properly? How can the factory prevent or fix this issue?

- *Possible Answers:*
  - The system might misplace candies or stop working.
  - Dust or poor lighting could interfere with sensor accuracy.
  - Using multiple sensors as a fail-safe could help.
  - Regular maintenance and calibration can ensure proper detection.

### Brainstorm Web

*(Encourage students to write or draw their ideas in a mind map.)*

**Main Idea:** Servo motor and sensor failures

Possible Problems: Motor failure, sensor inaccuracy, environmental factors

Solutions: Backup motors, multiple sensors, alerts, maintenance

### Strategies for Delivery

#### Strategy: "Think-Pair-Share"

- **Think:** Students write down their thoughts.
- **Pair:** Discuss with a partner.

- **Share:** Present ideas to the class, leading to a group discussion.

#### **Strategy: “Problem-Solution Chart”**

- Divide the board into two columns: Problem and Solution.
- List student-identified problems and brainstorm solutions together.

#### **Strategy: “Real-World Connection”**

- Show videos or pictures of real conveyor belt systems.
- Ask: *“Where have you seen sorting machines before?”* (Airports, factories, recycling centers, etc.)

### **3. Explore:**

#### **Teacher Role**

- Guide students in setting up the circuit using the color sensor and servo motor.
- Facilitate discussions on challenges in controlling multiple servo motors independently (e.g., power limitations, coding complexity).
- Encourage troubleshooting and problem-solving when students face errors in connections or programming.
- Ensure safety when handling electrical components.

#### **Expected From Students**

- Follow step-by-step instructions to connect the servo motor and color sensor on the Arduino board.
- Write and upload code to control the servo motor based on the sensor’s input.
- Identify challenges in controlling multiple servo motors independently and propose solutions.
- Watch the linked video for additional guidance.

#### **Parts of the Books Included**

# Explore



Connect the servo motor with the color sensor on the Arduino board.

What challenges might we face in controlling multiple servomotors independently?



## Activity - Do It

### Materials Needed

- Arduino board
- Color sensor (e.g., TCS3200)
- Servo motor
- Jumper wires
- Breadboard
- Computer with Arduino IDE

### Step-by-Step Instructions

#### 1. Connect the Color Sensor

- Attach the VCC and GND of the color sensor to the 5V and GND on the Arduino.
- Connect the sensor's S0, S1, S2, and S3 pins to the Arduino digital pins (e.g., D4, D5, D6, D7).
- The OUT pin of the sensor goes to an Arduino digital input (e.g., D8).

#### 2. Connect the Servo Motor

- The servo's red wire goes to 5V, the black wire to GND, and the yellow (signal) wire to D9 on the Arduino.

#### 3. Upload the Code

- Open the Arduino IDE and write a basic script to move the servo based on the detected color.

- Example: If the sensor detects red, the servo moves to position 1; if it detects blue, it moves to position 2.

#### **4. Test the Circuit**

- Place different colored objects in front of the sensor and observe the servo motor's movement.
- Adjust the code if necessary to refine accuracy.

#### **Reflection Questions**

- What happens when the sensor detects different colors?
- What difficulties did you encounter in wiring or coding the circuit?
- How could we control multiple servo motors without interference?
- What improvements can be made to the system for real-world applications?

#### **4. Explain:**

##### **Teacher Role**

- Explain how servo motors operate, including their internal components (DC motor, control circuit, potentiometer).
- Guide students to connect what they learn in the video with their hands-on experience.
- Ask questions to encourage critical thinking about real-world applications of servo motors.
- Facilitate a post-video discussion to reinforce key concepts.

##### **Expected From Students**

- Actively watch and take notes on key ideas from the video.
- Identify and explain the function of each part of the servo motor.
- Relate video content to previous hands-on experiments with servo motors.
- Share their observations and insights during class discussions.

##### **Parts of the Books Included**



## Watch It: Video-Based Learning

### Pre-Watching: Setting a Purpose

**Strategy:** *Prediction & Inquiry*

#### Implementation:

- Teacher asks:  
*What do you already know about how servo motors move?*  
*What controls their movement, and how precise do you think they are?*
- Students write predictions or discuss in small groups before watching.

### During Watching: Active Engagement

**Strategy:** *Guided Observation*

#### Implementation:

- Students focus on:
  - *What are the main components of a servo motor?*
  - *How does the potentiometer help in positioning?*
  - *What role does the control circuit play?*
- Students take notes on how signals determine movement and how servo motors differ from other motors.

### Post-Watching: Discussion & Reflection

**Strategy:** *Socratic Seminar & Hands-on Connection*

#### Implementation:

- Teacher asks open-ended questions to spark discussion:
  - *Why do we need precise control in servo motors?*
  - *How does the servo maintain accuracy in positioning?*

- *Can you think of real-life examples where servo motors are essential?*
- Students connect the video's information to their Arduino servo experiments.
- The teacher challenges students to identify possible issues (e.g., inaccurate positioning, lag, overheating) and propose solutions.

## 5. Elaborate:

### Teacher Role

- Encourage students to apply their knowledge by building a circuit with multiple servo motors.
- Guide them through troubleshooting power issues, signal interference, and coding challenges.
- Foster creativity by challenging students to modify the circuit for real-world applications.
- Provide hints instead of direct answers to promote problem-solving skills.

### Expected From Students

- Connect and program multiple servo motors on an Arduino board.
- Ensure proper wiring and power distribution for smooth motor operation.
- Modify the code to control each servo motor independently.
- Troubleshoot issues related to power, signal conflicts, or movement accuracy.

### Parts of the Books Included



### Practice Task: Attaching Multiple Servo Motors

#### Materials Needed

- Arduino board
- 2 servo motors

- Color sensor
- Breadboard
- Jumper wires
- External 5V power source
- Arduino IDE

## Step-by-Step Instructions

### 1. Connect the Servo Motors to Power & Ground:

- Connect the red wire of all servos to the 5V pin (or an external power source if using more than two servos).
- Connect the brown wire of all servos to the GND pin.

### 2. Connect the Signal Pins to Arduino:

- Connect the orange wire of each servo to different digital PWM pins on the Arduino (e.g., D9, D10)

### 3. Write and Upload the Code:

- Open Arduino IDE and use the Servo.h library.
- Write a program to control multiple servos independently.
- Example code snippet:

```
#include <Servo.h>
```

```
Servo servo1;
```

```
Servo servo2;
```

```
Servo servo3;
```

```
void setup() {
```

```
servo1.attach(9); // First servo on pin 9
```

```
servo2.attach(10); // Second servo on pin 10
```



```
servo3.attach(11); // Third servo on pin 11  
}
```

```
void loop() {  
  
servo1.write(0);  
  
servo2.write(90);  
  
servo3.write(180);  
  
delay(1000);
```

```
  
servo1.write(180);  
  
servo2.write(90);  
  
servo3.write(0);  
  
delay(1000);  
}
```

#### 4. Test and Troubleshoot the Setup:

- If servos twitch or move unpredictably, check power supply.
- If some servos don't respond, verify the signal wiring.
- Try modifying the code to make the servos move in a pattern.

#### Reflection Questions

- *Why do we need an external power supply for multiple servo motors?*  
**Suggested Answer:** Servos draw a lot of current, and the Arduino's built-in 5V pin may not provide enough power.
- *What challenges did you face while controlling multiple servos? How did you fix them?*  
**Suggested Answer:** Some servos didn't move smoothly due to power issues. I used an external 5V power supply to stabilize them.
- *How can multiple servos be used in real-world automation systems?*  
**Suggested Answer:** They can be used in robotic arms, conveyor belts, and automated sorting machines.

## 6. Evaluate:

### Teacher Role

- Facilitate the assessment process by observing students' projects and their integration of the color sensor.
- Provide constructive feedback on both the technical implementation and design choices.
- Encourage students to reflect on their learning journey and problem-solving strategies.
- Use a rubric-based evaluation to ensure consistency and fairness in grading.

### Expected From Students

- Demonstrate how the color sensor is integrated into their project.
- Explain how the sensor detects colors and how the system responds.
- Update the Engineering Design Process (EDP) paper with the newly added sensor.
- Reflect on challenges faced and how they were solved.

### Parts of the Books Included



### Showcase: Evaluating the Final Project

Students will present their working model showcasing the color sensor in action. They will also update their EDP paper, describing:

- How the color sensor was wired and programmed
- What problems they encountered and how they solved them
- How the sensor improves the project's functionality

### Rubric for Assessment

| Criteria       | Excellent (4)   | Good (3)         | Needs Improvement (2) | Incomplete (1)    |
|----------------|-----------------|------------------|-----------------------|-------------------|
| Integration of | Sensor is fully | Sensor works but | Sensor is             | Sensor is missing |

|                                                       |                                                                            |                                                                |                                                           |                                           |
|-------------------------------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------|-----------------------------------------------------------|-------------------------------------------|
| <b>Color Sensor</b>                                   | functional, properly wired, and improves the project's purpose.            | needs minor improvements.                                      | connected but does not function properly.                 | or not connected correctly.               |
| <b>Code Implementation</b>                            | Code is optimized, well-commented, and runs without errors.                | Code works but is not optimized.                               | Code has errors and needs debugging.                      | No functional code is implemented.        |
| <b>Engineering Design Process (EDP) Documentation</b> | EDP paper is fully updated with diagrams, explanations, and test results.  | EDP paper is updated but lacks some details.                   | EDP paper is partially updated with minimal explanations. | No updates to the EDP paper.              |
| <b>Presentation &amp; Explanation</b>                 | Clear, confident explanation of how the sensor works and why it was added. | Explanation is clear but lacks detail in sensor functionality. | Some explanation, but missing key details.                | Unable to explain how the sensor works.   |
| <b>Problem-Solving &amp; Reflection</b>               | Identified challenges, provided solutions, and reflected on improvements.  | Identified challenges but solutions are unclear.               | Few challenges discussed with limited solutions.          | No discussion of challenges or solutions. |

#### **Relation to Standards**

This project aligns with:

#### **NGSS (Next Generation Science Standards)**

- *MS-ETS1-2*: Evaluate competing design solutions using a systematic process.
- *MS-PS4-3*: Integrate qualitative scientific and technical information to support claims.

#### **ISTE Standards for Students**

- *Innovative Designer*: Students use a variety of technologies to solve problems.
- *Computational Thinker*: Students develop and test solutions to improve a system.

## Color Sorting Machine

### Learning Outcomes

By the end of the theme student will be able to:

- Select Optimal Cam Type.
- Identify the best belt type for your system.
- Research and apply methods to prevent jams.
- Ensure smooth object flow through the system.
- Identify Belt Drive Components: Recognize pulleys, shafts, and belts.
- Create conveyors and bins for sorting.
- Assemble System: Accurately build the sorting system.
- Create Linkages: Design linkages to control sorting arms or gates.
- Wire Color Sensor: Connect a color sensor to an electrical circuit.
- Control Multiple Servos: Connect and synchronize multiple servo motors.
- Calibrate Sensor: Test and fine-tune the color sensor.
- Power and Sync Servos: Configure the circuit to power and sync servos.
- Learn New Functions: Understand and apply new functions and loops.
- Link Signals: Connect sensor input to servo motor output.
- Integrate Arduino with Python: Interface Arduino with Python for control.

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# Lesson 6

UNIT 02: [Advanced]

Chapter 01: Robotics

Lesson 01: Show the results

Time Frame: 90 minutes

| Activity Name                     | Duration | Description                                                                                                                                                                                                      |
|-----------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Engage: The Mystery Display       | 15 min   | The teacher presents a malfunctioning device with no visible output and asks students how they would know what’s happening inside. This leads to a discussion on the importance of visual feedback using an LCD. |
| Think: Brainstorming the Solution | 15 min   | Students brainstorm different ways to display information in an automated system. The class discusses the role of LCDs in real-world applications.                                                               |
| Explain: How LCDs Work            | 20 min   | The teacher explains the structure of an LCD, its communication with Arduino, and how it can display messages and sensor data. Students examine an LCD and                                                       |

|                                                   |        |                                                                                                                                                                            |
|---------------------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                   |        | connect it to an Arduino.                                                                                                                                                  |
| <b>Practice:<br/>Programming the<br/>LCD</b>      | 30 min | Students follow step-by-step instructions to write and upload code that displays messages on the LCD. They test their setups and troubleshoot errors.                      |
| <b>Elaborate:<br/>Customizing the<br/>Display</b> | 25 min | Students modify their code to display sorting bin statuses (e.g., "Sorting Plastic", "Sorting Metal"). They apply conditional logic based on sensor inputs.                |
| <b>Evaluate:<br/>Showcase &amp;<br/>Feedback</b>  | 15 min | Students demonstrate their working LCD integration, explaining how their code works. The teacher assesses their work using a rubric and provides feedback for improvement. |

## 1. Engage:

### Teacher Role:

- Facilitate a discussion about how machines detect and display information.
- Ask thought-provoking questions to encourage critical thinking.
- Guide students in brainstorming how an LCD screen could be used in a color-sorting machine.
- Use real-world examples to help students relate to the concept.

### Expected From Students:

- Actively participate in the discussion by sharing their ideas.
- Reflect on how machines identify and display colors.
- Predict possible challenges a color-sorting system might face.
- Engage in a hands-on activity using colored objects to simulate a sorting system.

### Intro Story: The Smart Candy Dispenser Challenge

#### Strategy: "Real-World Connection"

#### Implementation:

1. Start with a question: *"Have you ever used a vending machine or a candy dispenser?"*
2. Introduce the story of the high-tech candy shop where customers select a candy, but the machine isn't displaying the correct color.
3. Ask:
  - *"What could be causing this problem?"*
  - *"How do you think machines detect and display colors?"*

4. Engage students in a hands-on demonstration:
  - Give each group colored candies or objects.
  - Ask them to think about how a sensor would detect and display the color on an LCD.
  - Have students share their ideas before revealing how a color sensor works.

### Story Reflection

#### Questions:

1. *Why is the LCD screen important in the candy sorting system?*
2. *What challenges could arise if the LCD does not update correctly?*
3. *How can we ensure that the correct color is displayed on the screen?*

#### Suggested Answers:

1. The LCD screen helps users see the detected color in real-time before dispensing the candy.
2. If the LCD does not update, the wrong color might be displayed, leading to customer confusion.
3. We can write an accurate program to ensure the color sensor and LCD communicate correctly.

### Strategy: "Think-Pair-Share"

#### Implementation:

1. **Think:** Students reflect on the questions individually.
2. **Pair:** They discuss their answers with a partner.
3. **Share:** Each pair shares their ideas with the class, promoting collaborative learning.

## 2. Think:

### Questions and Possible Answers

*"What if we needed to see the whole process from the control room? Which component should we add?"*

#### Possible Answers:

- LCD Screen – Displays real-time sorting data near the machine.
- Computer Interface – Sends data to a screen in the control room.
- Wireless Module (Wi-Fi/Bluetooth) – Transmits sorting data remotely.
- Camera Module – Streams a live video feed of the sorting process.

#### Brainstorm Web:

- Why do we need remote monitoring? → To ensure the system is working correctly.
- What components can help? → LCD, LED indicators, Wi-Fi, camera.
- Which is the best solution? → LCD for nearby monitoring, Wi-Fi/camera for distant monitoring.

### Teaching Strategies:

#### 1. "Think-Pair-Share"

**Implementation:**

1. Ask students to think individually about the best component.
2. Pair them up to discuss their ideas.
3. Share their answers with the class and justify their choices.

**2. "Predict the Problem"****Implementation:**

- Ask: *"What if we had no way to monitor the process? What problems could arise?"*
- Have students list challenges and propose solutions.

**3. "Decision Matrix"****Implementation:**

- Give students a table with different monitoring components (LCD, Wi-Fi, Camera).
- Ask them to rate each option based on cost, efficiency, and ease of integration.
- Discuss the best solution for the sorting machine.

### 3. Explain:

**Teacher Role:**

- Explain how an LCD screen works in displaying sensor data.
- Guide discussions on how the LCD enhances the color sorting machine.
- Ask students to analyze the information shown on the LCD in real-world applications.

**Expected From Students:**

- Engage actively in discussions.
- Share observations from the video.
- Explain how an LCD helps monitor sorting results.
- Connect the LCD's role to other real-life automation systems.

**Parts of the Books Included:**



# Explore



Watch It: [video link](#)

## Pre-Watching (Set a Purpose)

### Goal:

- Understand how an LCD works in real-time data display.
- Identify the connections between the color sensor, microcontroller, and LCD.

### Strategy:

- Use a KWL Chart (*Know, Want to Know, Learned*).
- Ask students: “What do you already know about LCDs?”

## During Watching (Active Engagement)

### Focus Points:

- How does the LCD receive data from the color sensor?
- What type of information does it display?
- How does this improve sorting efficiency?

### Strategy:

- Guided Notes: Provide fill-in-the-blank sheets to track key points.
- Pause & Predict: Stop at key moments and ask, “What do you think will happen next?”

## Post-Watching (Discussion & Reflection)

### Discussion Questions:

1. How does the LCD display different colors detected by the sensor?
2. Why is the LCD useful in real-time monitoring?
3. How could we improve the information shown on the LCD?

### Strategy:

- Think-Pair-Share: Students discuss answers with a partner before sharing.
- Draw It Out: Have students sketch a simple LCD screen with labeled data.

## 4. Elaborate:

### Teacher Role:

- Encourage creativity and experimentation.
- Guide students in connecting the LCD to their circuit.
- Assist with troubleshooting and debugging code.
- Foster discussions on how this enhancement improves the system.

### Expected From Students:

- Assemble and connect the LCD to the color sensor system.
- Write and test the Arduino code for displaying sensor data.
- Observe the real-time sorting results on the LCD.
- Reflect on improvements and document them in the EDP paper.

### Parts of the Books Included:



### Practice Steps

#### Materials Needed:

- Arduino board
- LCD display (I2C or standard)
- Color sensor module
- Servo motors
- Jumper wires
- Breadboard
- Computer with Arduino IDE

### Step-by-Step Instructions:

#### 1. Set Up the Hardware:

- Connect the LCD display to the Arduino using I2C/SDA-SCL (A4 and A5) or direct pin wiring.
- Ensure the color sensor is properly wired to the microcontroller.
- Ensure the servo motors is properly wired to the microcontroller.

#### 2. Write & Upload the Code:

- Open the Arduino IDE and write a program to read RGB values from the sensor.
- Display the detected color name or RGB values on the LCD.
- Upload and run the code.

#### 3. Test & Debug:

- Place different colored objects in front of the sensor.
- Observe if the LCD correctly displays the detected color.
- Adjust the code if necessary for better readability and accuracy.

#### 4. Enhance & Document:

- Modify the display to show sorting decisions (e.g., *"Red Object → Bin 1"*).
- Update the EDP Paper: Explain the impact of adding the LCD.

#### 5. Final Step:

**Students finalize their projects, document all improvements, and present their functioning LCD integration to the class.**

### Reflection Questions:

- *What data is displayed on the LCD, and why is it useful?*
- *What challenges did you face while coding or wiring the LCD?*
- *How does this enhancement improve the user experience?*

### Strategy for Reflection:

- Think-Pair-Share: Discuss challenges and improvements in pairs before sharing with the class.
- Sketch & Explain: Students draw their circuit and annotate how data flows.

## 5. Evaluate:

### Teacher Role:

- Facilitate student presentations and demonstrations.
- Assess students' ability to program and integrate the LCD.
- Provide constructive feedback and guidance for improvements.
- Use a rubric to ensure clear evaluation criteria.

#### Expected From Students:

- Demonstrate their working LCD system and explain how it functions.
- Answer assessment questions related to LCD programming and integration.
- Reflect on challenges faced and improvements made.
- Update their EDP paper with the final design modifications.

#### Parts of the Books Included:



#### Showcase & Evaluation

##### Evaluation Criteria (Rubric):

| Criteria                   | Excellent (3 pts)                                             | Good (2 pts)                                                          | Needs Improvement (1 pt)                        |
|----------------------------|---------------------------------------------------------------|-----------------------------------------------------------------------|-------------------------------------------------|
| <b>LCD Integration</b>     | LCD correctly displays data and updates dynamically.          | LCD works but may have minor issues.                                  | LCD does not display data correctly.            |
| <b>Code Functionality</b>  | Code is well-structured, efficient, and commented.            | Code runs with minor inefficiencies or lacks comments.                | Code has major errors or is incomplete.         |
| <b>Project Explanation</b> | Student clearly explains LCD integration and troubleshooting. | Student provides a basic explanation but lacks clarity in some areas. | Student struggles to explain LCD functionality. |
| <b>EDP Documentation</b>   | EDP paper reflects thorough testing and improvements.         | EDP paper includes some modifications but lacks depth.                | EDP paper is incomplete or missing updates.     |

#### Relation to Standards:

##### NGSS (Next Generation Science Standards):

- **MS-ETS1-3:** Analyze data from tests to determine how best to improve a solution.

- **MS-PS3-3:** Apply scientific principles to design a system that converts energy from one form to another.

**Common Core (Mathematics & ELA):**

- **CCSS.MATH.PRACTICE.MP1:** Make sense of problems and persevere in solving them.
- **CCSS.ELA-LITERACY.SL.8.1:** Engage in collaborative discussions about project improvements.